ECE 3210 hw07

1. These are old exam questions. They are easy to check numerically, so you should verify your work with Python or NumPy. However, you must still be able to do these calculations by hand on an exam.

   (a) Given the discrete sequences

   $$f = \{1, 1, 1, 1, 0, 0, 0\}, \qquad g = \{1, 2, 3, 4, 3, 2, 1\},$$

   find the 7-point circular convolution of f and g. Represent your answer as a vector or array.

   Solution: The linear convolution is

   $$f * g = [1, 3, 6, 10, 12, 12, 10, 6, 3, 1, 0, 0, 0].$$

   Folding its samples modulo 7 gives

   $$y[n] = (f \circledast g)[n] = (f * g)[n] + (f * g)[n + 7] \quad (0 \le n \le 6),$$

   where out-of-range samples are zero. Thus

   $$y = [1 + 6,\ 3 + 3,\ 6 + 1,\ 10,\ 12,\ 12,\ 10] = [7, 6, 7, 10, 12, 12, 10].$$

   (b) Using the DFT, verify your answer by computing the FFT of both sequences, multiplying them, and reporting the inverse FFT result. Do this using the real part only if the result is complex-valued.

   Solution: Using NumPy, compute the two 7-point DFTs, multiply them pointwise, and take the inverse DFT:

   ```
   import numpy as np

   F = np.fft.fft([1, 1, 1, 1, 0, 0, 0])
   G = np.fft.fft([1, 2, 3, 4, 3, 2, 1])
   y_fft = np.fft.ifft(F * G).real
   np.rint(y_fft).astype(int)
   ```

   Rounding removes only floating-point roundoff, giving

   $$y_{\text{FFT}} = [7, 6, 7, 10, 12, 12, 10],$$

   in agreement with the direct circular convolution.

(c) What length of circular convolution is required to compute the linear convolution of f and g? Express your answer as a single integer.

Solution: For sequences of lengths L_f and L_g, a circular convolution of length at least $L_f + L_g - 1$ preserves the full linear convolution. Here both sequences have length 7, so

$$N = L_f + L_g - 1 = 7 + 7 - 1 = 13.$$

2. Consider the following sequences:

$$f_k = \{-1, 0, 1, -1, 0, 1\}, \qquad g_k = \{1, 1, 1, 1, 1, 1\}.$$

(a) Find the linear convolution $y_{k,\text{lin}}$ of f_k and g_k.

Solution: Each output sample is

$$y_{k,\text{lin}} = \sum_{m=0}^{5} f_m g_{k-m},$$

with f_m and g_m treated as zero outside $0 \leq m \leq 5$. Evaluating the overlap at each shift gives

$$y_{\text{lin}} = [-1, -1, 0, -1, -1, 0, 1, 1, 0, 1, 1].$$

(b) Now find the circular convolution of f_k and g_k.

Solution: Folding the linear-convolution result into six samples is

$$y_{\text{circ}} = [0, 0, 0, 0, 0, 0].$$

3. Consider the discrete-time signal

$$x[n] = \begin{cases} 1 + \cos\left(\frac{\pi n}{4}\right) - 0.5\cos\left(\frac{3\pi n}{4}\right), & 0 \le n \le 7, \\ 0, & \text{otherwise.} \end{cases}$$

Compute the 8-point DFT $X_8[k]$ of $x[n]$, and plot $X_8[k]$ for $0 \le k \le 7$.

Solution: Use the inverse DFT synthesis formula to match the complex-exponential terms in $x[n]$ to the DFT coefficients:

$$x[n] = \frac{1}{8}\sum_{k=0}^{7} X_8[k]e^{j2\pi kn/8}.$$

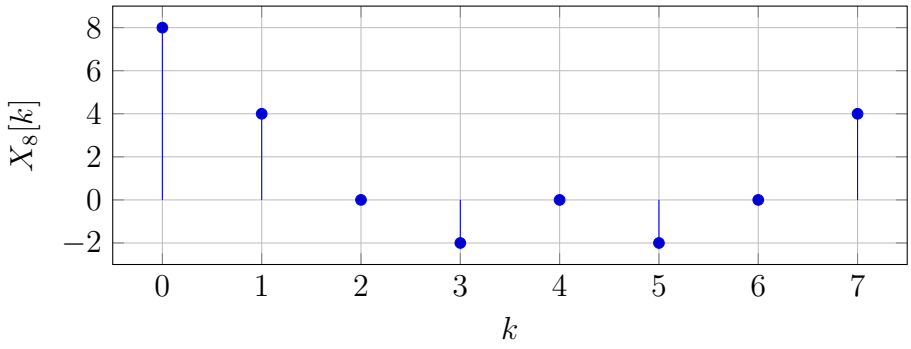
Expanding the cosines gives

$$\begin{aligned} x[n] &= 1 + \frac{1}{2}e^{j2\pi n/8} + \frac{1}{2}e^{-j2\pi n/8} - \frac{1}{4}e^{j2\pi(3)n/8} - \frac{1}{4}e^{-j2\pi(3)n/8} \\ &= \frac{1}{8}\left(8 + 4e^{j2\pi n/8} + 4e^{j2\pi(7)n/8} - 2e^{j2\pi(3)n/8} - 2e^{j2\pi(5)n/8}\right). \end{aligned}$$

Matching coefficients with the synthesis formula gives $X_8[0] = 8$, $X_8[1] = X_8[7] = 4$, $X_8[3] = X_8[5] = -2$, and $X_8[2] = X_8[4] = X_8[6] = 0$. Therefore

$$X_8[k] = [8, 4, 0, -2, 0, -2, 0, 4].$$

The discrete plot of these DFT values is:



4. (a) Suppose you have a signal $x(t)$ that you sample at 200 Hz to obtain a 20-point sequence x_k. If you take a 20-point DFT of this sequence, what frequencies are represented by the DFT output? Give your answer in Hz.

Solution: The frequency spacing of an N-point DFT is f_s/N. With $f_s = 200$ Hz and $N = 20$,

$$f_k = \frac{kf_s}{N} = 10k \text{ Hz}, \qquad k = 0, \ldots, 19.$$

Therefore the represented frequencies are

$$\{0, 10, 20, 30, \ldots, 190\} \text{ Hz}.$$

(b) Now suppose you zero-pad this sequence x_k to have 30 samples and take a 30-point DFT. What frequencies are represented in that DFT? Give your answer in Hz.

Solution: Zero-padding changes the DFT length to $N = 30$, so the bin spacing is

$$\Delta f = \frac{f_s}{N} = \frac{200}{30} = \frac{20}{3} \text{ Hz}.$$

Hence

$$f_k = \frac{20k}{3} \text{ Hz}, \qquad k = 0, \ldots, 29,$$

or, rounded to two decimal places,

$$\{0, 6.67, 13.33, 20, 26.67, \ldots, 193.33\} \text{ Hz}.$$